Role of free holes in nBTI degradation in GaN-on-Si MOS-channel HEMTs

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***Abstract*— To understand the peculiar kinetic of the negative Bias Temperature Instabilities (nBTI) in GaN-on-Si MOS-channel HEMTs (MOSc-HEMTs), temperature (T), gate bias ($V_{G,stress}$) and modification of the epitaxial structure are used to determine if free holes are responsible for long-term threshold voltage (Vth) shift. Furthermore, UV assisted nBTI measurements were carried out to stimulate free holes generation in the epitaxy structure near the gate. TCAD simulations validate the assumption that nBTI is a multifactorial degradation mixing free holes capture in the Al_2O_3 gate oxide and C_N deep acceptor charge state modification in the buried GaN:C layer.**

Keywords—GaN-on-Si HEMT, nBTI, UV assisted nBTI, Carbon in GaN.

I. Introduction

Wide bandgap devices such as silicon carbide (SiC) and gallium nitride (GaN) are emerging as an alternative to the traditional silicon based transistors. GaN devices have shown numerous advantages for power applications especially for medium power (<2kW) and for multi-MHz applications and are breaking through the mass market using several technologies to reach normally-OFF behavior [1-3]. One possibility lies on using a fully recessed MOS gate on unintentionally doped GaN (GaN:UID) as presented in Fig. 1 (called MOS-channel HEMT). Lately, excellent interface properties have been demonstrated using Al_2O_3 based gate oxide [4] even after gate recess etching process. Still, GaN devices reliability remains a hot topic through the hot electrons on-resistance (R_{ON}) related degradation or the threshold voltage instabilities (ΔV_{TH}) under harsh conditions [5-7]. These parameters degradations can affect the transistor lifetime and its dynamic performance during operation. Recently a strong effort focused on Bias Temperature Instability (BTI) helped to catch the interplay between epitaxy and nBTI [5] or pBTI [6-7] degradations. Indeed, carbon doped layer (GaN:C) tends to induce an additional V_{TH} drift mechanism through C_N deep acceptor traps (E_A = 0.8eV) charge state modification. Nevertheless, no actual free holes related nBTI degradation was demonstrated so far. This study presents a thorough analysis of nBTI degradation through temperature and UV assisted measurements supported by TCAD simulations to understand the role of free holes in nBTI degradation of MOSc-HEMTs.

II. Device description

200mm GaN-on-Si wafers were processed using MOCVD (metalorganic chemical vapor deposition) directly on p-type Si substrate as presented in Fig. 1. First, a thick "transition layer" for strain management and a carbon doped GaN layer (GaN:C), that ensures a vertical and lateral electrical insulation and improves GaN breakdown voltage, are deposited. An unintentionally doped GaN layer (GaN:UID) is then deposited to form the conductive channel. An AlN spacer followed by a top AlGaN barrier creates the 2D electron gas (2DEG) at the GaN/AlGaN interface. In-situ SiN based passivation is used to stabilize AlGaN surface.

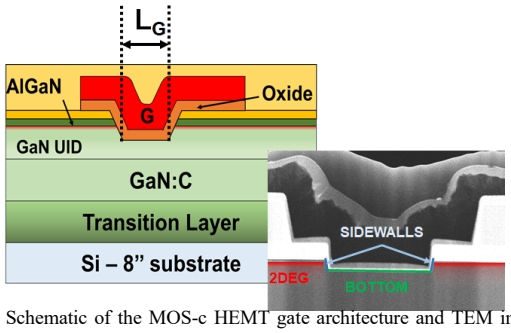

Fig.1: Schematic of the MOS-c HEMT gate architecture and TEM image showing fully recessed gate 2D structure including sidewalls, corners and gate bottom.

ICP RIE etching is used to create the GaN cavity where a 30nm ALD Al_2O_3 is deposited to form the MOS stack on GaN:UID (Fig. 1) including bottom, corners and sidewalls. Two natures of Al_2O_3 will be analyzed (type A or B). Tested transistor have a 1µm gate length (L_G) and a gate width (W_G) of 200µm or 8000µm. Three different epitaxial stacks will be addressed in this study (Fig. 2) with (a) a standard MOSc-HEMT epitaxy as presented here above, (b) a 1µm thick GaN:UID layer and (c) a 1µm thick p-GaN layer (N_A^- ~ 1e17 cm^{-3}). These stacks are measured for constant device configuration and gate process to understand the impact of epitaxy on nBTI degradation

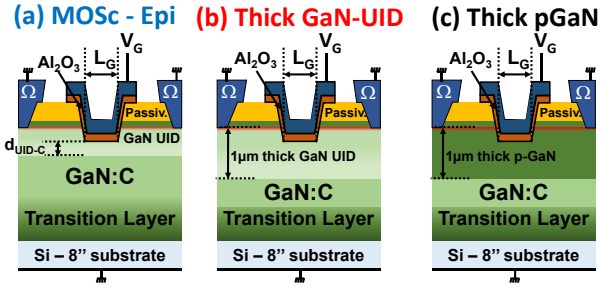

Fig. 2: Schematic of the three epitaxial stacks used in this study. (a) Regular epitaxy structure for MOSc-HEMT integration where d_{UID-C} represents the distance between GaN:C layer and gate bottom, (b) a 1µm thick GaN-UID channel and (c) a 1µm thick p-GaN layer (N_A^- ~ 1e17 cm^{-3}).

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III. NBTI RESULTS

Fig. 3 presents the ultra-fast nBTI measurement as used in [7]. This setup enables to capture V_{TH} evolution using MSM (Measurement Stress Measurement) procedure with sub-10µs $I_D(V_G)$ ramps where V_{TH} is extracted at constant current $I_{D,TH} = 5.10^{-8}.L_G/W_G$. Stress voltage is set at $V_{G,Stress}$ = -5V during Stress phase (unless stated otherwise) and at $V_{G,Stress}$ = 0V during Relaxation phase. During the whole

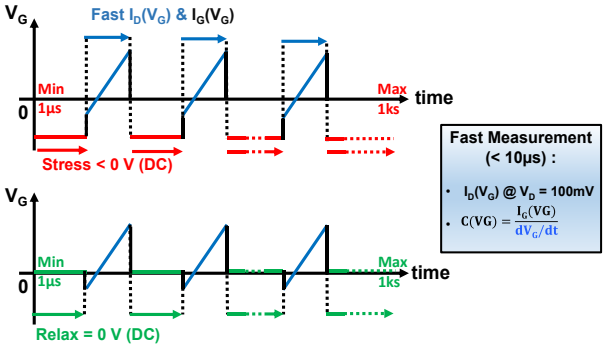

Fig. 3: Principle of the ultra-fast nBTI measurement using WGFMU. Fast ramps (<10µs) are designed to minimize measurement impact on stress/recovery sequences. Threshold voltage Vth is monitored through IV or CV characteristics [7].

procedure, V_D is set to 0V to get a symmetrical electrical field along the channel.

Temperature dependent measurements are presented in Fig. 4 on standard MOSc-HEMT epitaxy. Unlike previous studies [5,6,8], high temperature nBTI degradation seems to be ruled by an energy, extracted through Arrhenius plot, around E_A = 0.36eV (compared to 0.8eV extracted previously [5]). This energy is consistent with the role of Carbon in GaN epitaxy where conduction of free holes in the volume of the buffer through a defect band has been recently considered [9, 10]. The impact of gate voltage amplitude $V_{G,Stress}$ on nBTI transients at 200°C for two types of Al_2O_3 (A and B) is presented on Fig. 5 and reveals that for low stress amplitude ($|V_{G,Stress}| < 2V$) only a small degradation

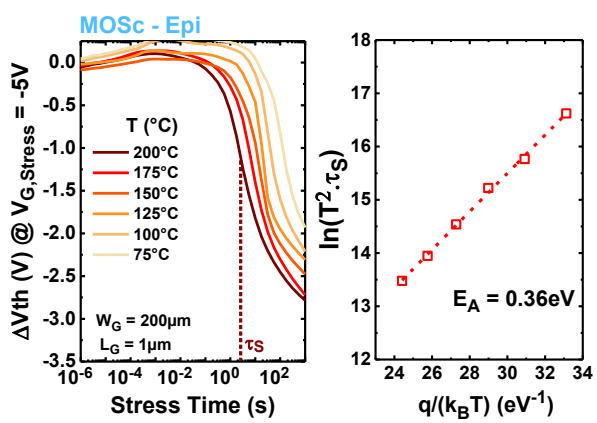

Fig. 4: (Left) Temperature dependent nBTI measurement on 200µm wide MOS-c HEMT (L_G = 1µm) at $V_{G,Stress}$ = -5V and (right) Arrhenius analysis of the time constant (τ_S) extracted through local maxima of ∂ΔVth/∂log(t). Energy of the high temperature nBTI strong degradation lies around E_A = 0.36eV as in [9].

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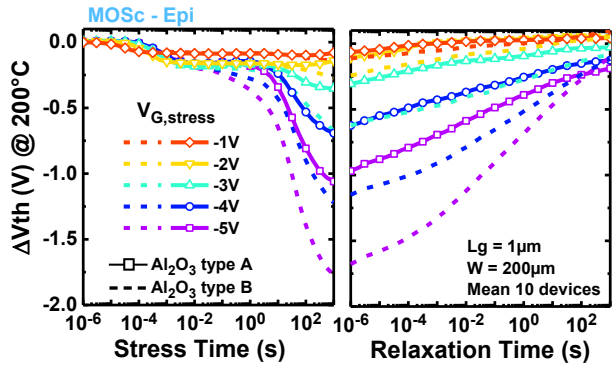

Fig. 5: (left) Stress and (right) recovery nBTI transients for $V_{G,Stress}$ from -1V to -5V at T = 200°C. $V_{G,Stress}$ and oxide deposition procedure (type A and B) impact long term degradation which confirms the actual free holes capture process inside pre-existing defects of Al_2O_3 can occur in GaN MOSc-HEMTs. First degradation at low stress time (related to GaN:C as in [5]) is nearly independent on $V_{G,Stress}$ or oxide type.

related to C_N charge state modification [5] is observed. The long-term degradation only starts to appear at higher stress amplitude ($|V_{G,Stress}| > 2V$) similarly to pBTI studies where electrons trapping occurs in Al_2O_3 pre-existing defects [6]. Furthermore, the nature of the gate oxide (type A or B) strongly affects the nBTI, which is a sign that long-term degradation relates to holes charging of gate oxide pre-existing defects.

Epitaxial structure also induces major modification of nBTI transient dynamics (Fig. 6) at {-5V; 200°C}. The GaN:C/channel distance (d_{UID-C}) induces a delay in the onset of the long-term degradation which clearly states the free hole transport towards the Al_2O_3/GaN:UID interface is epitaxy-dependent through a complex RC network around the gate. This "epi-effect" is also visible on Fig. 7 where different [C] in GaN:C layer modify the kinetics of the nBTI transients at 200°C. GaN:C layer is consequently the major provider of free holes in the structure which is fully consistent with recent studies [9-11]. Similar high temperature nBTI measurements were performed on thick GaN:UID and thick p-GaN configurations (Fig. 8). pGaN thick layer can provide a lot of free holes which directly triggers the nBTI phenomena even at short stress time whereas thick

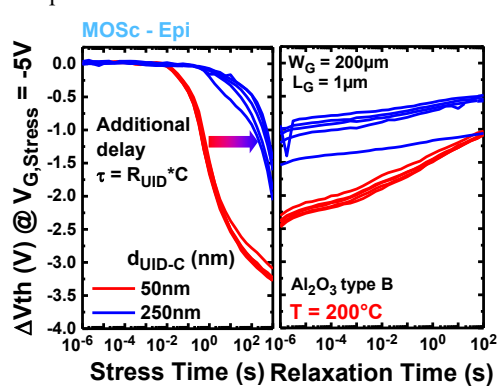

Fig. 6: (left) Stress and (right) recovery nBTI transients for $V_{G,Stress}$ = -5V at 200°C as a function of distance between GaN:C and gate bottom (d_{UID-C}) on Al_2O_3 type B. Strong influence of the GaN:UID layer underneath the gate is observed showing the nBTI phenomena kinetics is epitaxy-dependent through the RC-network surrounding the gate

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GaN:UID severely delays the transport of free holes towards the gate oxide even at 200°C and therefore the trapping in oxide defects.

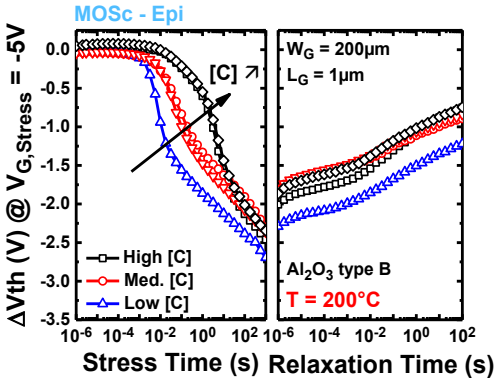

Fig. 7: (left) Stress and (right) recovery nBTI transients for $V_{G,Stress}$ = -5V at 200°C as a function of [C] in GaN:C layer. The higher the [C] the longer the delay for nBTI to trigger in the Al_2O_3 gate oxide. This demonstrates that nBTI kinetics in GaN MOSc-HEMT relies also on GaN:C which is the sole source of free holes in the material [8,9,10].

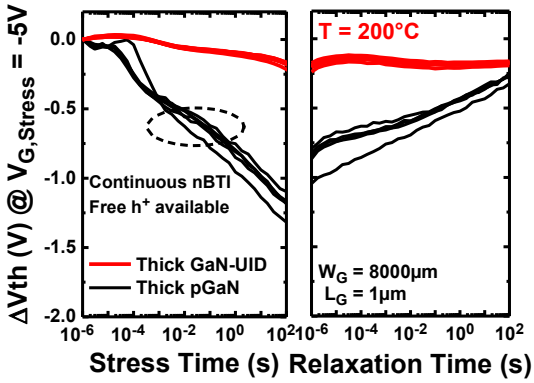

Fig. 8: (left) Stress and (right) recovery nBTI transients for $V_{G,Stress}$ = -5V at 200°C for 1µm thick GaN-UID and 1µm thick p-type GaN. The nBTI kinetics of the thick pGaN samples depicts the presence of free holes available directly at the beginning of the stress whereas the use of thick GaN-UID tends to delay severely the apparition of holes trapping in Al_2O_3.

To assess the role of free holes in the long-term degradation of GaN MOSc-HEMT, light-assisted nBTI MSM procedure was applied on the thick GaN:UID epitaxy at 25°C (Fig. 9).

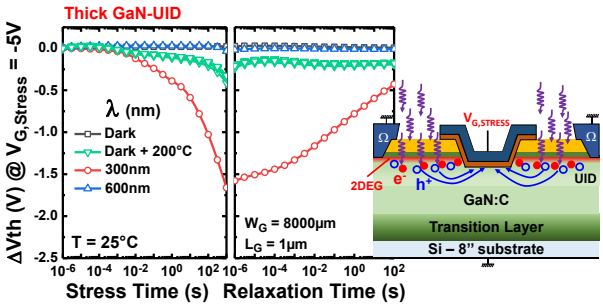

Fig. 9: Influence of illumination during nBTI measurements on 1µm thick GaN-UID multi-fingers MOSc-HEMT. At 25°C for dark and 600nm illumination, no variation is observed whereas using UV illumination (λ < 350nm) creates free holes in the GaN-UID layer which can be easily trapped in Al_2O_3 at $V_{G,Stress}$ = -5V.

It clearly states that under dark or visible light ($\lambda > hc/E_G$), no degradation is observed whereas the use of UV light ($\lambda \leq hc/E$) stimulates the electron-hole pair creation during stress which enables the capture of generated free holes inside the Al_2O_3 pre-existing defects thanks to the negative bias on the gate.

A precise analysis of the 200°C nBTI transient of MOSc-HEMT (Fig. 10) reveals a specific acceleration of the drift around t4 ≈ 5s. Complementary, the associated fast $I_D(V_G)$ recorded at different stress time (t1 to t5) shows that a subthreshold slope (SS) degradation can be observed when the nBTI acceleration is triggered around t4.

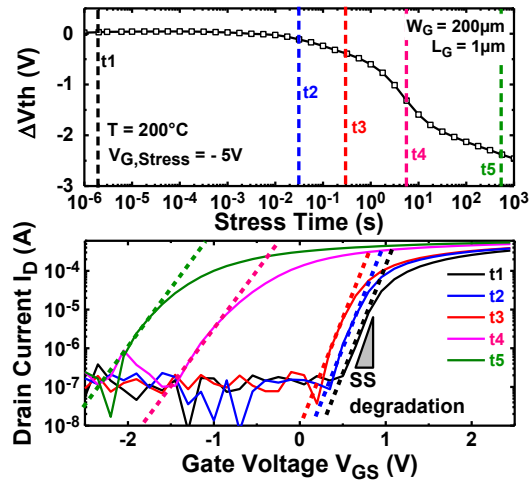

Fig. 10: Comparison of fast $I_D(V_G)$ (bottom) extracted at multiple time during stress at $V_{G,Stress}$ = -5V (top). The first part of nBTI (t1 to t3) only shifts the characteristics whereas a SS degradation is observed at t4 and after.

This loss of electrostatic control under the gate is reversible and not due to an interface degradation but to a modification of the back-gating effect linked to the proximity of the GaN:C layer which presents a strong capacitive coupling with the electron channel through the in-between fully depleted GaN:UID layer.

IV. TCAD SIMULATIONS

To understand further the degradation process, TCAD simulations were used to calculate the properties of an epitaxial stack during a "so-called" nBTI stress. The simulated structure consists of a MOSc-HEMT full structure including Al_2O_3 gate oxide and field plates on a GaN:UID layer over a thick 2e18 cm^{-3} C-doped GaN layer ($E_T - E_V$ = 0.8eV) with a 50% donor compensation ($E_C - E_T$ = 0.1eV) and finally the transition layer over a Si p-type substrate. Fig. 11 depicts the chronogram of the simulation which relies on the calculation of an equilibrium state and the application of a -5V gate stress during 1ks. Band structures and electrical parameters were recorded at equilibrium, 1µs after the stress application, at the end of the stress, 1µs after the stress and at the end of recovery.

A strong change in ionized C_N traps ("-" → "0") is visible under the gate. This modification of charge also induces a reduction in the back gating effect produced by the ionization of the C_N. This effect can produce a loss in electrostatic

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control of the channel electron charge and thus alter the V_{TH} and the SS parameters just after the stress phase in the case of Fig. 11.

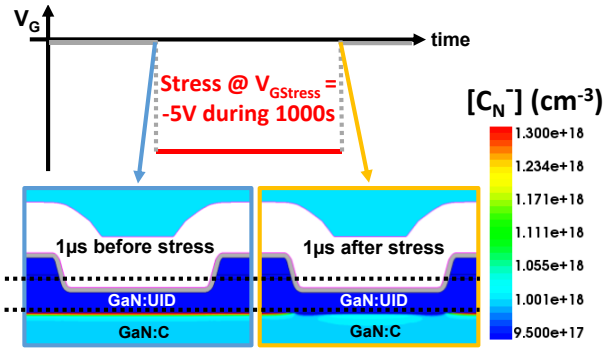

Fig. 11: TCAD simulation (top) of the impact of negative $V_{G,Stress}$ ionized $[C_N]$ acceptors ($E_T = E_V + 0.8$eV) in the GaN:C layer. A strong deionization of C_N ("-"→0) acceptors is observed 1µs after the 1ks stress at -5V on the gate. The change in ionized C_N acceptors can strongly affect the electrostatic control underneath the gate and, thus the subthreshold slope.

Ionized $[C_N]$ ("-") and free holes density $[h^+]$ are plotted along the depth of the simulated structure from the gate oxide to the volume of the transition layer on Fig. 12. Equilibrium state shows a full ionization of the C_N at the interface between GaN:UID and GaN:C induced by Fermi energy levels alignment as well as a very reduced free hole density around $[h^+] \approx 10^9$ cm^{-3} in the GaN:C volume (consistent with [11]). Plus, at the end of the {-5V; 1ks} stress a strong $[h^+]$ is observed at the gate oxide/GaN:UID interface as well as a strong diminution of the ionized C_N traps concentration. These aspects correlate with the experimental observations presented in the previous section.

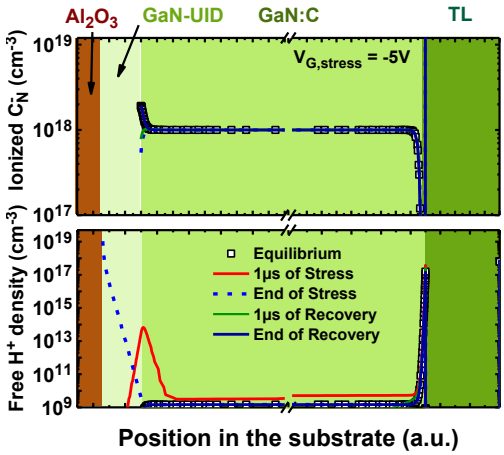

Fig. 12: Comparison of Ionized C_N (top) and free holes density (bottom) in the structure at different time during stress and recovery. Deionization of C_N and strong holes concentration $[h^+]$ underneath the gate at 1ks of stress are observed. It shows that nBTI is a multifactorial degradation as pBTI [6].

The nBTI degradation process in GaN MOSc-HEMTs is thus multifactorial with a (i) C_N traps charge state modification underneath the gate ("-" → "0") at the GaN:UID/GaN:C interface and (ii) free holes trapping inside Al_2O_3 gate oxide pre-existing defects.

Conclusion

This paper provides an in-depth study of the nBTI degradation in GaN-on-Si fully recessed gate MOSc-HEMTs. Temperature, gate voltage and epitaxial structure influence on nBTI have been thoroughly investigated using ultra-fast MSM procedure. TCAD simulations have provided strong insights on the presence of free holes in the structure which can be highly concentrated at oxide/GaN:UID interface. nBTI in GaN MOSc-HEMT is modeled as a combination of C_N traps deionization under the gate in the GaN:C layer and of free holes trapping in Al_2O_3 pre-existing defects.

References

[1] P. Moens et al., "An industrial process for 650V rated GaN-on-Si power devices using in-situ SiN as a gate dielectric," 2014 IEEE 26th International Symposium on Power Semiconductor Devices & IC's (ISPSD), 2014, pp. 374-377, doi: 10.1109/ISPSD.2014.6856054.

[2] J. Ren et al., "Experimental characterization of the fully integrated Si-GaN cascoded FET," 2018 IEEE 30th International Symposium on Power Semiconductor Devices and ICs (ISPSD), 2018, pp. 216-219, doi: 10.1109/ISPSD.2018.8393641.

[3] S. Yang, Z. Zheng, L. Zhang, W. Song and K. J. Chen, "GaN MIS-HEMTs With Surface Reinforcement for Suppressed Hot-Electron-Induced Degradation," in IEEE Electron Device Letters, vol. 42, no. 4, pp. 489-492, April 2021, doi: 10.1109/LED.2021.3057933.

[4] W. Vandendaele et al., "A Novel Insight on Interface Traps Density (Dit) Extraction in GaN-on-Si MOS-c HEMTs," 2020 IEEE International Electron Devices Meeting (IEDM), 2020, pp. 23.5.1-23.5.4, doi: 10.1109/IEDM13553.2020.9371965.

[5] A. G. Viey et al., "Investigation of nBTI degradation on GaN-on-Si E-mode MOSc-HEMT," 2019 IEEE International Electron Devices Meeting (IEDM), 2019, pp. 4.3.1-4.3.4, doi: 10.1109/IEDM19573.2019.8993588.

[6] A. G. Viey et al., "Carbon-related pBTI degradation mechanisms in GaN-on-Si E-mode MOSc-HEMT," 2020 IEEE International Electron Devices Meeting (IEDM), 2020, pp. 23.6.1-23.6.4, doi: 10.1109/IEDM13553.2020.9371938.

[7] A. G. Viey et al., "Study on the difference between ID(VG) and C(VG) pBTI shifts in GaN-on-Si E-mode MOSc-HEMT," 2021 IEEE International Reliability Physics Symposium (IRPS), 2021, pp. 1-8, doi: 10.1109/IRPS46558.2021.9405221.

[8] N. Zagni et al., "“Hole Redistribution” Model Explaining the Thermally Activated RON Stress/Recovery Transients in Carbon-Doped AlGaN/GaN Power MIS-HEMTs," in IEEE Transactions on Electron Devices, vol. 68, no. 2, pp. 697-703, Feb. 2021, doi: 10.1109/TED.2020.3045683.

[9] Nardo, A., et al. "Positive and negative charge trapping GaN HEMTs: Interplay between thermal emission and transport-limited processes." Microelectronics Reliability 126 (2021): 114255. doi: 10.1016/j.microrel.2021.114255

[10] C. Koller, G. Pobegen, C. Ostermaier and D. Pogany, "Effect of Carbon Doping on Charging/Discharging Dynamics and Leakage Behavior of Carbon-Doped GaN," in IEEE Transactions on Electron Devices, vol. 65, no. 12, pp. 5314-5321, Dec. 2018, doi: 10.1109/TED.2018.2872552.

[11] M. J. Uren et al., "“Leaky Dielectric” Model for the Suppression of Dynamic RON in Carbon-Doped AlGaN/GaN HEMTs," in IEEE Transactions on Electron Devices, vol. 64, no. 7, pp. 2826-2834, July 2017, doi: 10.1109/TED.2017.2706090.

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